

  FACULDADE DE CIÊNCIAS UNIVERSIDADE DO PORTO   FACULDADE DE ENGENHARIA UNIVERSIDADE DO PORTO	Master Degree in Artificial Intelligence Artificial Intelligence and Society	2024/2025 1 st Year 1 st Semester
INSTRUCTOR: Miriam Seoane Santos		

Individual Assignment 4: Missing Data

Submission Deadline: Jan 26

Objective

The purpose of this assignment is to study the problem of missing data in machine learning tasks and explore strategies for its characterization and mitigation. **The deliverables of this assignment are a 2-page report with your insights and a jupyter notebook (or Python project) with your experiments. The assignment might be subjected to defense.**

Instructions

In this individual assignment, you should investigate an open-ended question regarding the problem of missing data using empirical or exploratory analysis.

Open-ended Projects

- A. Develop an empirical analysis on the impact of missing data in machine learning algorithms. You may start by collecting several datasets without missing values and implement some solutions to create different missing mechanisms (MCAR, MAR, and MNAR). Then, select among several approaches to handle missing data (e.g., classifying with missing values, performing data imputation) and compare the obtained classification performance across different approaches. Do the results vary depending on the missing mechanism? Is the top performer in terms of classification metrics also the top performer if the focus shifts towards Predictive or Distributional Accuracy?
- B. Develop an exploratory study to generate some hypotheses on how to diagnose missing data mechanisms. Current methods to handle missing data rely on certain assumptions about missing mechanisms and not all strategies are suitable for all mechanisms. For that reason, it is important to correctly determine the underlying missing mechanism in data to apply the most appropriate strategies. In this project, your goal is to explore, propose, and implement some visualization strategies to intuitively allow users to understand the missing data patterns and mechanisms arising in their datasets. You can start by collecting complete datasets and generate different missing data mechanisms in your data to evaluate your proposed visualizations. Then, choose 1/2 real-world datasets comprising missing values. Is your approach able to highlight some trends and patterns of the missing data?

Deliverables

For this assignment, the following deliverables are required:

- A report (up to 2 pages -- using [Template 1](#) or [Template 2](#)) summarizing your main findings. The report should include the following sections:
 - **Introduction:** Brief description of the problem definition and objectives;
 - **Methodology:** Explain the rationale of your approach and the steps taken;
 - **Results and Discussion:** In-depth discussion of your findings;
 - **Conclusions:** Summarize key findings, limitations, and learnings;
 - **References:** Cite all resources used in your work (e.g., books, research papers, websites, tools, packages). Use a consistent citation format.
- A Jupyter Notebook (or .py project) containing the complete code used for analysis, inline comments explaining important aspects and visualizations/data included in the report.

Evaluation Criteria

Your submission will be evaluated based on the following criteria:

- **Creativity and Problem Definition:** Ability to define a clear and well-motivated question for investigation; originality in framing the approach to the analysis; creativity in the used tools.
- **Quality of Experiments:** Quality and thoroughness of the experiments designed to address the problem; appropriate use of meta-information to draw conclusions; correct implementation of methods, and clear motivation and explanation of approaches.
- **Reflections and Insights:** Depth of analysis and critical reflection on the obtained results; ability to interpret the implications of meta-information on model performance, preprocessing, or other aspects; discussion of limitations and challenges.
- **Clarity and Organization of Deliverables:** Writing and logical flow and structure of the report and the Jupyter Notebook (or project).
- **Bibliography:** Quality of the sources and tools and their relevance to the analysis.

Bibliography

- Van Buuren, Stef., [Flexible imputation of missing data](#). CRC press, 2018.
- [Imputation of Missing Values](#), scikit-learn documentation.
- [missingno](#): missing data visualization module for Python
- [mdatagen](#): a Python library for the generation of Artificial Missing Data
- [naniar](#): Data Structures, Summaries, and Visualizations for Missing Data
- **Public Data Repositories:**
 - [Kaggle Datasets](#)
 - [KEEL Datasets](#)
 - [UCI Machine Learning Repository](#)
 - [data.world](#)
 - [Open ML](#) (Bischl, B., Casalicchio, G., Feurer, M., Gijbbers, P., Hutter, F., Lang, M., ... & Vanschoren, J. (2017). [Openml benchmarking suites](#). *arXiv preprint arXiv:1708.03731*.)